Pricing a Chokepoint: A Synthetic-Control Estimate of the Brent Crude Premium from the 2026 Strait of Hormuz Disruption

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Abstract

Roughly a fifth of the world's seaborne crude passes through the Strait of Hormuz, so its disruption on 28 February 2026 was widely expected to move global oil prices. Measuring that effect, however, is not straightforward: As Brent crude represents a unique global benchmark, there is no untreated control unit available from which a counterfactual price can be directly inferred. We construct that missing counterfactual with the *synthetic control method* (SCM), building a price path from a pool of 19 macro-financial donor series (metals, agriculturals, equities, foreign exchange, rates, and volatility) chosen to share Brent's exposure to common global factors while staying off the oil-supply causal path. To keep the result from hinging on any single modelling choice, the synthetic is formed as the median of an ensemble of five estimators with different inductive biases. We benchmark the framework using the Russian invasion of Ukraine on 24 February 2022, which represents the closest comparable historical oil supply shock. We then apply the framework to estimate the effects of a disruption in the Strait of Hormuz. Over the post-event window to mid-June 2026 we find an average Brent premium of roughly 56 % (ensemble median; interquartile range 51 % to 57 %), an estimate that is robust to placebo-based inference and a range of robustness and contamination checks. We frame the estimate as a reduced-form, data-driven complement to the structural scenario estimates produced by energy institutions, and discuss its use as an input to strategic-reserve, monetary, and fiscal decisions.

Contents

1 Introduction

The Strait of Hormuz is the narrowest binding constraint in the global oil system. Roughly one-fifth of world petroleum-liquids consumption transits the lane between Iran and Oman, carrying crude from Saudi Arabia, Iraq, the United Arab Emirates, Kuwait, Iran, and Qatar to seaborne markets. What makes the chokepoint so consequential is the absence of a redundant route. The Saudi East–West and UAE Habshan–Fujairah pipelines together divert only a small fraction of this volume overland, and no maritime bypass exists, so the overwhelming majority of Gulf crude has nowhere else to go. This leaves Hormuz the single strongest bottleneck in the world's oil-transport network. The lane has seen periodic near-misses, among them the 1980s tanker war, the 2019 Abqaiq strike, and recurring tensions through the mid-2020s, each producing a transient risk-premium spike but never an actual closure. The 28 February 2026 disruption is the first realisation of that long-anticipated tail risk.

Because crude oil is an input to almost every industry and a cost borne across nearly all economies, a disruption on this scale was always going to move prices, and Brent duly rose sharply in the days that followed. We study Brent rather than another benchmark for two reasons. While the major crude indices are highly correlated and the choice between them is largely immaterial, Brent is the reference price against which the majority of internationally traded crude is settled, and we use its spot quotation rather than a futures contract. Yet the raw price jump is uninformative on its own. It conflates the disruption with everything else moving markets at the same time, and it says nothing about how the effect evolves. This motivates a causal question:

> *What is the causal effect of the 2026 Strait of Hormuz disruption on the price of Brent crude over a policy-relevant post-event window?*

We deliberately target the *total* causal effect rather than any single transmission channel, and we estimate it as a *path* over the post-event window rather than a single number. The applications drive this choice. Emergency strategic-reserve releases are sized against an implied price displacement. The monetary-policy decision to look through an oil shock or to tighten turns on its size and persistence. Fiscal stabilisation in subsidising oil importers scales with the move. And sovereign hedging programmes price disruption scenarios against exactly such magnitudes. Each lever acts on the trajectory of the effect, not on a peak-day figure. The object we need is therefore the average gap between observed Brent and the price that would have prevailed absent the disruption, traced over the whole window. Recovering that counterfactual path, which has no directly observable analogue, is the methodological core of this paper.

Standard observational designs cannot deliver it. Brent is a single, globally traded benchmark, which rules out difference-in-differences (no panel of comparable treated and control units), regression discontinuity (no running variable around a cutoff), and instrumental variables (no credible instrument for a one-off geopolitical event). The *synthetic control method* (Abadie et al., 2010; Abadie, 2021) is built for exactly this situation. It forms a weighted basket of donor series whose pre-event co-movement with the outcome is high and projects that basket forward as the counterfactual. It yields a full counterfactual path rather than a point estimate, makes its identifying assumptions auditable through the donor pool, and requires no structural stance on whether the move is supply, risk premium, or expectations, since its estimand is a reduced-form total effect.

The tools most commonly used to analyze such price shocks include event studies, structural VARs, and institutional scenario models. However, they tend to address narrower or different questions. Event studies measure abnormal returns in a short window around the event but yield no multi-month path and miss persistence. Structural VARs (Kilian, 2009; Baumeister and Hamilton, 2019) decompose prices into supply, demand, and inventory shocks but produce no counterfactual path and rely on identification restrictions with no settled precedent for a chokepoint event. And institutional scenario models propagate an assumed barrel shortfall through a supply–demand model, so their counterfactual is imposed rather than read from observed prices. We see these as complements rather than competitors. Our cross-asset, data-driven estimate and a structural scenario estimate answer different questions, the price gap implied by observed co-movement versus the price implied by an assumed shortfall, so their agreement is itself informative, and in practice the two can be used side by side.

Methodologically, our closest neighbour is the SCM-on-price design of Becker et al. (2021). That work studies a regulated retail-fuel market with cross-country donor units. Neither is available here, since Brent is a global crude benchmark and no untreated country or market can serve as a donor. We therefore replace geographic donors with cross-asset donors, including metals, agriculturals, equities, foreign exchange, rates, and volatility, and apply the design under an exogenous geopolitical treatment. To our knowledge this combination has not been used for any previous oil shock, so the contribution is as much about establishing whether such a design is valid in this setting as about the number it produces.

A single estimate, however, cannot be told apart from an artefact of the donor pool, pre-window, model class, and regularisation. We therefore benchmark the design on the nearest comparable historical oil-supply shock, the 2022 Russian invasion of Ukraine (Brent \$95→\$130 within two weeks), and check that the same machinery recovers that documented magnitude before estimating the Hormuz effect afresh on its own pre-window. We do not transfer the Russia fit to Hormuz. The two events sit in very different market regimes, and we estimate each one independently. Because both are fit on a common 19-series donor pool, we can additionally compare the donor weights across regimes as a check on the stability of the factor structure SCM relies on.

To prevent any single modelling choice from driving the result, the synthetic is formed as the median of an ensemble of five estimators with different inductive biases: convex SCM, augmented SCM, elastic net, gradient-boosted trees, and Bayesian ridge. Fitting a synthetic that tracks pre-event Brent closely is the easy part. The substantive work of this paper is the validation that challenges the donor pool and the no-interference assumption it rests on, namely placebo-based inference, leave-one-donor-out checks, structural-break tests, a signed contamination-bias analysis, and a comparison against naïve univariate counterfactuals. Our contribution is thus twofold: a data-driven Brent-price ATT for the Hormuz disruption that complements existing structural estimates, and a transparent template for validating cross-asset synthetic control under a single-unit geopolitical treatment.

The remainder of the paper reviews the relevant literature (Section 2), sets out the factor model, estimand, and identifying assumptions (Section 3), describes the data and donor audit (Section 4), details the empirical strategy and validation battery (Section 5), and reports results for both events (Section 6) before concluding (Section 7).

2 Related Literature

We organise the review around problems rather than individual papers, because our contribution sits at the intersection of three otherwise separate strands: how causal counterfactuals are built for a single aggregate unit, how the oil-price effect of geopolitical shocks is conventionally measured, and how inference is conducted with one treated unit and a short post-period. Tools borrowed from machine learning and time series are introduced where they are used (Section 5 and the appendices). Here we position the causal design.

2.1 Building counterfactuals for a single aggregate unit

The problem of constructing a counterfactual for one treated unit is what synthetic control was invented to solve. The method originates with Abadie and Gardeazabal (2003) and Abadie et al. (2010), who build the counterfactual as a convex combination of untreated donors matched on pre-treatment outcomes and formalise permutation (placebo) inference for the single-unit case. Abadie et al. (2015) extend it to comparative politics, and Abadie (2021) codifies the practical requirements we adopt: a long, shock-free pre-window, donors clean of the treatment (the SCM analogue of SUTVA), and a good pre-treatment fit as a precondition for credibility. A subsequent literature relaxes the convex-hull restriction. The augmented SCM of Ben-Michael et al. (2021) adds a ridge bias-correction permitting limited extrapolation when the treated unit lies outside the donor hull, Doudchenko and Imbens (2016) embed SCM, difference-in-differences, and regression in a common regularised-regression family that allows unconstrained weights, and Athey et al. (2021) recast the problem as matrix completion. A caution runs through all of it. Convex SCM weights are *non-unique* when donors are correlated (Abadie and L'Hour, 2021), so many weight vectors yield near-identical fit and individual weights should not be over-interpreted. We use this entire toolbox by running an ensemble that spans the convex, augmented, and regularised-regression cases, and we read cross-model weight dispersion through the non-uniqueness result rather than as instability.

2.2 Where SCM has been applied, and the gap we occupy

Despite this methodological maturity, SCM has rarely been applied to a daily market price. Most peer-reviewed applications estimate effects on *non-price* aggregates such as GDP, trade, and employment, including under discrete geopolitical shocks. Gharehgozli (2017) estimates the GDP cost of sanctions on Iran and Mancini et al. (2024) the trade effect of the 2022 Russia–Ukraine war. The closest mechanical precedent to a price application is Becker et al. (2021), who apply SCM to Austrian retail fuel prices with other European countries as donors. That design differs from ours in three ways that matter, as its treatment is an endogenous regulation rather than an exogenous shock, its outcome an administered retail price rather than a traded benchmark, and its donors cross-country rather than cross-asset. Juárez et al. (2024) use an oil-price shock as the treatment but on regional GDP, not on a price. The intersection we occupy, which is SCM on a crude-benchmark *price* under an exogenous *geopolitical* disruption with cross-asset donors, has no direct precedent, and that is the gap this thesis addresses. Because the Hormuz event is recent, the only existing analysis is institutional (Kiel Institute for the World Economy, 2026; Oxford Institute for Energy Studies, 2026; World Bank, 2026; International Energy Agency, 2026; Federal Reserve Bank of Dallas, 2026). These share one structure, namely to assume a barrels-per-day shortfall, propagate it through a structural model, and output a price path, and none applies synthetic control to the observed price. They form the comparison set against which we position, and complement, our estimate.

2.3 Oil prices and geopolitical risk: the incumbent methods

The methods conventionally used for oil-price effects answer narrower or different questions than ours. The dominant approach is the structural VAR with identified supply, demand, and inventory shocks (Kilian, 2009; Baumeister and Hamilton, 2019), with Hamilton (2009) the narrative anchor for 2007–08. These decompose price changes into structural components but yield no counterfactual path and require identification restrictions for which a chokepoint event offers no standard precedent. A natural alternative would regress oil prices on the Geopolitical Risk index of Caldara and Iacoviello (2022), but this recovers the *average* response to a unit of geopolitical risk pooled across many events, a different object from the effect of one specific disruption, and the index spikes endogenously to the very event we study, so it can serve neither as our estimand nor as a clean donor. Short-window event studies (Brown and Warner, 1985) complete the incumbent set and give a magnitude but not the multi-month path policy requires.

2.4 Inference with one treated unit and a short post-period

Our setting makes inference non-standard, and the recent literature counsels restraint in two places. The first is inference itself. Permutation and placebo tests (Abadie et al., 2010, 2015) are the field standard, with validity resting on an approximate-symmetry assumption (Canay et al., 2017). Hahn and Shi (2017) show that for convex SCM this effectively requires Gaussian errors, which bounds how far placebo p-values can be pushed for our nonlinear ensemble members, and Chen and Yan (2023) formalise the in-time placebo into a mixed test with a permutation p-value. The second is pretesting. Roth (2022) shows that pre-trend tests have low power at applied sample sizes and can manufacture false confidence, and Rambachan and Roth (2023) propose bounding rather than testing such violations, so we accordingly adopt a report-and-interpret stance toward pre-period diagnostics rather than imposing pass/fail thresholds. Finally, the value of a donor depends on how reliably it tracks the common factor. A literature on matching reliability before differencing shows that a good pre-period fit can *mask* rather than remove contamination, the logic we use to split donors by pre-window factor R^2 when signing the contamination bias (Section 6). Multiple testing across donors is controlled by Benjamini–Hochberg (Benjamini and Hochberg, 1995).

3 Conceptual Framework

This section sets out the data-generating process we assume for Brent, derives the estimand and identifying assumptions from it, and explains why an *ensemble* of estimators is the appropriate proxy when the functional form of the counterfactual is unknown. The factor model in Section 3.1 is the organising device. It makes donor selection an economic exercise, renders each identifying assumption transparent, and gives the contamination bias an explicit sign.

3.1 A factor model of Brent

We decompose log Brent into a component spanned by global macro-financial factors common to many assets and an oil-specific component:

$$p_t^{\text{Brent}} = \underbrace{\lambda' F_t}_{\text{common global factors}} + \underbrace{o_t}_{\text{oil-specific (supply, chokepoint, geopolitics)}} + \varepsilon_t. \tag{1}$$

F_t collects the five drivers that move Brent in normal times, namely the global industrial cycle, the US dollar, real interest rates, global risk appetite, and China demand, each proxied by non-oil assets. The oil-specific term o_t is the object of interest, the supply and chokepoint component a Hormuz disruption shifts. Each donor j admits its own factor representation $X_{jt} = \gamma_j' F_t + u_{jt}$ with, by construction, *zero loading on* o_t. A clean donor is exposed to the same global factors as Brent but is not physically routed through oil supply. A synthetic control is then a weighted donor basket $\hat{p}_t = \sum_j w_j X_{jt}$, and the logic of the method follows directly from (1). If the weights reproduce Brent's factor loadings, $\sum_j w_j \gamma_j \approx \lambda$, the basket tracks Brent through the pre-window, where o_t is small and stable. When a chokepoint shock fires, o_t jumps for Brent but not for the basket, the two decouple, and that decoupling *is* the estimated premium. Donor selection is therefore the economic exercise of spanning F_t while excluding o_t, that is, choosing identification over fit. This is precisely why the energy complex (WTI, refined products, petrocurrencies), despite the highest raw correlation with Brent, is excluded by design. It loads on the very o_t we are trying to measure.

3.2 Potential outcomes and the estimand

Equation (1) describes the observed price, while the causal question concerns the price that would have prevailed without the event. Let Y_t^N be Brent's potential log-price absent the disruption and Y_t^I under it, with treatment beginning at T_0. The treated-unit effect is $\tau_t = Y_t^I - Y_t^N$ for $t \geq T_0$. With only one treated unit, Y_t^N is never observed after T_0. The synthetic supplies the missing term, $\hat{Y}_t^N = \sum_j w_j X_{jt}$, and the estimated effect is the gap $\hat{\tau}_t = Y_t^I - \hat{Y}_t^N$, reported in percent as $\text{gap}_t = 100\,[\exp(Y_t^I - \hat{Y}_t^N) - 1]$. We summarise the post-window by the *average treatment effect on the treated*,

$$\widehat{\text{ATT}} \;=\; \frac{1}{|W|} \sum_{t \in [T_0,\, T_0+W]} \text{gap}_t. \tag{2}$$

We prefer the ATT to the peak or terminal gap for two reasons. It integrates over the entire path, including any reversion, which is the object that reserve sizing, monetary pass-through, and fiscal buffering act on, and averaging dampens single-day jumps from liquidity or futures-roll mechanics.

3.3 Identifying assumptions

Because the synthetic stands in for an unobserved counterfactual, the estimate is only as credible as the assumptions that license it, and the factor model makes each one explicit and checkable.

1. **Common-factor spanning (pre-period fit).** The donor basket must reproduce Brent's factor loadings, so that good pre-event tracking reflects shared F_t rather than coincidence. Checked by pre-period RMSPE, walk-forward cross-validation, and moment matching.
2. **Donor cleanliness (SUTVA / no interference).** No donor may carry the treatment. The loading γ_j must not load on o_t, and donor j must not be hit through a channel specific to the focal event. Defended by an a-priori audit and confirmed by model-agnostic break tests at T_0.
3. **Regime stability.** The loadings λ, γ_j must hold across the pre-window and the projection window, so that a basket which tracked Brent before T_0 remains a valid counterfactual after it. Checked by distance-correlation and structural-break diagnostics, and, across events, by comparing donor weights between regimes.

Assumptions 1 and 3 are testable from observed data, whereas assumption 2 is the genuinely untestable one, since donor cleanliness concerns the unobserved X_{jt}^N. The next subsection shows how the factor model nonetheless lets us sign the *direction* in which a violation of assumption 2 would move the estimate.

3.4 The direction of contamination bias

The factor representation turns the informal worry that "the donors might be affected by the war" into a quantity with a definite sign. Writing $\delta_{jt} = X_{jt} - X_{jt}^N$ for the event-induced component of donor j, the estimation error is, as an exact identity,

$$\hat{\tau}_t - \tau_t \;=\; Y_t^N - \hat{Y}_t^N \;=\; -\sum_j w_j\, \delta_{jt}. \tag{3}$$

Table 1: The five-estimator ensemble. Each contributes a distinct inductive bias, and the headline is the cross-model median gap.

Estimator	Inductive bias	Reference
Convex SCM	Non-negative donor weights summing to one, synthetic inside the donor convex hull	Abadie et al. (2010)
Augmented SCM	Convex SCM plus ridge bias-correction, limited extrapolation outside the hull	Ben-Michael et al. (2021)
Elastic net	Signed, sparse+ridge regression, lasso term drops donors entirely	Doudchenko and Imbens (2016)
Gradient boosting (XGBoost)	Nonlinear, tree-based, captures interactions, cannot extrapolate beyond the training range	Chen and Guestrin (2016)
Bayesian ridge	Bayesian linear regression on donor levels (regression component of BSTS), conjugate Gaussian–Gamma prior	Brodersen et al. (2015)

The bias is *minus* the weight-averaged donor contamination. Its sign is therefore an empirical matter rather than an assumption. If contaminated donors are pushed *up* (oil-linked terms-of-trade, inflation→rates), the synthetic is pulled up and the ATT is *under*-estimated, which is conservative. If they are pushed *down* (demand crowd-out, risk-off), the ATT is over-estimated. The identity (3) is exact, but δ_{jt} is itself unobserved, so signing the bias in practice requires estimating each δ_{jt} against a Brent-free reference, a deliberately naïve construction whose assumptions and limits we set out, and apply, in Section 6. What the theory delivers here is the *structure*, which is which way a given contamination pushes the estimate, not a certified magnitude. The same identity also distinguishes two regimes of contamination by timing. *Contemporaneous* contamination occurs at T_0 and is screened by the audit and break tests. *Delayed* contamination accrues over the post-window, enters through the projection, biases the gap toward zero, and grows with horizon, so a gap that *declines* as the window lengthens is its signature, and the short-horizon gap is the least-contaminated estimate.

3.5 Why an ensemble

A single specification cannot be trusted because the counterfactual's functional form is unknown. Equation (1) need not be linear in the donors, and the convex-hull restriction of classical SCM may or may not bind. Rather than commit to one form, we proxy the counterfactual with five estimators whose inductive biases bracket the plausible cases (Table 1): a simplex-constrained average, a ridge-extrapolating correction, a sparse signed regression, a nonlinear tree ensemble, and a Bayesian linear model. An estimated premium that survives across biases this different is far more likely to reflect a real effect than a single-model artefact. We report the cross-model *median* as the headline, which is robust to any one model misbehaving, and the inter-quartile range as a model-uncertainty band that complements each model's own within-model interval. A note on labelling: the fifth estimator is a Bayesian *ridge* regression, not full Bayesian structural time series. It implements only the regression-on-donors component $\beta' x_t$ of Brodersen et al. (2015) and omits the trend, seasonal, and spike-and-slab selection components. At our sample size, and with no strong weekly seasonality, the regression term carries most of the signal, but the model is best read as a Bayesian-flavoured diversifier rather than a state-space causal-impact estimator (Appendix B).

4 Data

This section describes the outcome series, the donor candidates and the per-event cleanliness audit that selects among them, and the sample windows over which each event is fit and projected. Two themes recur and both follow from the factor model of Section 3. Donor selection is an identification choice rather than a fit-maximising one, and the length of the estimation window is itself a SUTVA consideration, not merely a sample-size one.

4.1 Outcome and sources

The outcome is the daily *Europe Brent Spot Price FOB* (EIA series RBRTEd), in USD per barrel, downloaded directly from the U.S. Energy Information Administration (U.S. Energy Information Administration, 2026). The donor candidates are 32 macro-financial series from Yahoo Finance (futures, ETFs, FX, rates, volatility), updated incrementally so the panel keeps pace with new data. Table 2 summarises the sources. All are public and redistributable, and the panel is rebuilt from source on each run so the analysis is reproducible against the latest available data (here, through 15 June 2026 for Brent, the binding constraint given the EIA's roughly one-week publication lag, and 23 June 2026 for the donors).

Table 2: Data sources. All series are daily and publicly redistributable.

Object	Source	Contents
Brent spot (outcome)	EIA, series RBRTEd	Europe Brent Spot FOB, USD/bbl, 1987–present
Donor candidates (32)	Yahoo Finance	Metals, agriculturals, equities, FX, rates, credit, volatility, crypto

4.2 The donor pool and the SUTVA audit

Donor selection is central to identification (Section 3). A series qualifies only if it (i) co-moves with Brent through common global factors, (ii) is *not* physically routed through the disruption being attributed, (iii) is liquid and daily-traded, and (iv) has sufficient pre-history. Criterion (ii) is the SUTVA (no-interference) condition and is enforced by a per-event audit that classifies each donor as Clean, Mild, or Heavy. For **Russia 2022**, 13 of 32 donors are flagged through identifiable channels, namely Russia/Ukraine grain exports (Wheat, Corn), palladium supply concentration, the EU energy-crisis pass-through to EUR and the dollar basket, and a sanctions-evasion narrative for Bitcoin. For **Hormuz 2026**, only Cotton is flagged (the indirect oil→polyester substitution channel), and no donor is physically routed through the Strait.

To make the two events directly comparable, we fit both on a single shared 19-donor pool, formed by retaining only the donors rated Clean for Russia, the more demanding of the two audits. This reduces the 32 candidates to 19, and because the Russia strict-clean set is itself a subset of the Hormuz clean set, every member stays clean for both events. Cotton, flagged Mild for Hormuz, is therefore excluded on the strength of the Russia audit rather than retained, so its indirect channel never enters the shared pool. One donor, VIX, is borderline. A full-history Bai–Perron test (Bai and Perron, 1998) flags a single break in its relationship with Brent on 2022-01-07, about 7 weeks before the invasion. We treat that as a loading-stability signal rather than a SUTVA violation, because it does not reproduce when the test is run on the analysis window alone, and a sensitivity check shows that keeping or dropping VIX moves the ensemble premium by less than 1 percentage point. We therefore retain VIX, which keeps the volatility factor represented. The energy complex (WTI, refined products, petrocurrencies) is excluded by construction. It has the highest raw correlation with Brent but loads on the very oil-specific shock we measure. Table 3 lists the 19-donor pool by factor category.

Table 3: The shared 19-donor pool, by factor category. The full 32-donor pool adds Russia-contaminated series (e.g. Wheat, Corn, Palladium, EUR, DXY, BTC, Copper, Iron Ore, US10Y) used only in the appendix robustness pool and as a contamination diagnostic.

Category	Donors (shared pool)
Metals (3)	Silver, Platinum, Gold
Agriculturals (3)	Coffee, Sugar, Live Cattle
Equities (2)	S&P 500, Nikkei 225
Foreign exchange (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN
Rates & credit (2)	TLT (long Treasuries), HYG (high-yield credit)
Volatility (1)	VIX (implied S&P 500 volatility)

4.3 Sample windows and summary statistics

The estimation window must be chosen, not maximised. For daily-traded benchmarks like Brent and its donors, a longer pre-history is not automatically better. These series pass through distinct market regimes, and a window that spans several of them risks teaching the synthetic a donor–Brent relationship that no longer holds in the projection period, a regime shift that re-enters as post-event interference and thus as a SUTVA violation. Our first task was therefore to locate the longest pre-window that contains *no* prior shock or regime break, rather than the longest pre-window available. Each event then has such a pre-window (for fitting) and a post-window (for projecting the counterfactual), each ∼20 months. Following Abadie (2021), the Russia pre-window starts 2020-07-01, deliberately *after* the March–May 2020 cluster (OPEC+ collapse, WTI-negative, acute COVID) that would otherwise impose an abnormal factor loading. The Russia post-window is truncated at 2022-09-30, the last business day before OPEC+'s October 2022 supply cut, a Brent-specific policy shock that would confound the invasion premium. The Hormuz post-window extends to the latest available data (2026-06-15). Table 4 reports summary statistics, and Figure 1 plots the full Brent series with both events.

Table 4: Summary statistics of Brent spot (USD/barrel) over each event sample. Pre/post split at T_0. Russia: T_0=2022-02-24; Hormuz: T_0=2026-02-28.

Event	Segment	n (days)	Mean	SD	Min	Max
	Full sample	572	75.95	24.76	36.33	133.18
Russia 2022	Pre-window	421	64.26	16.39	36.33	101.66
	Post-window	151	108.54	11.37	82.55	133.18
	Full sample	515	76.95	14.30	59.93	138.21
Hormuz 2026	Pre-window	443	72.08	6.58	59.93	88.66
	Post-window	72	106.92	12.31	77.24	138.21

The two pre-windows differ in a way that matters later. The Russia pre-period is volatile (SD \$16.4, spanning the COVID recovery and the 2021 reflation rally), while the Hormuz pre-period is much calmer (SD \$6.6). This difference drives much of the contrast in the two events' inference, because a volatile pre-period widens the placebo distribution and weakens Russia's permutation test even though its magnitude is well-documented.

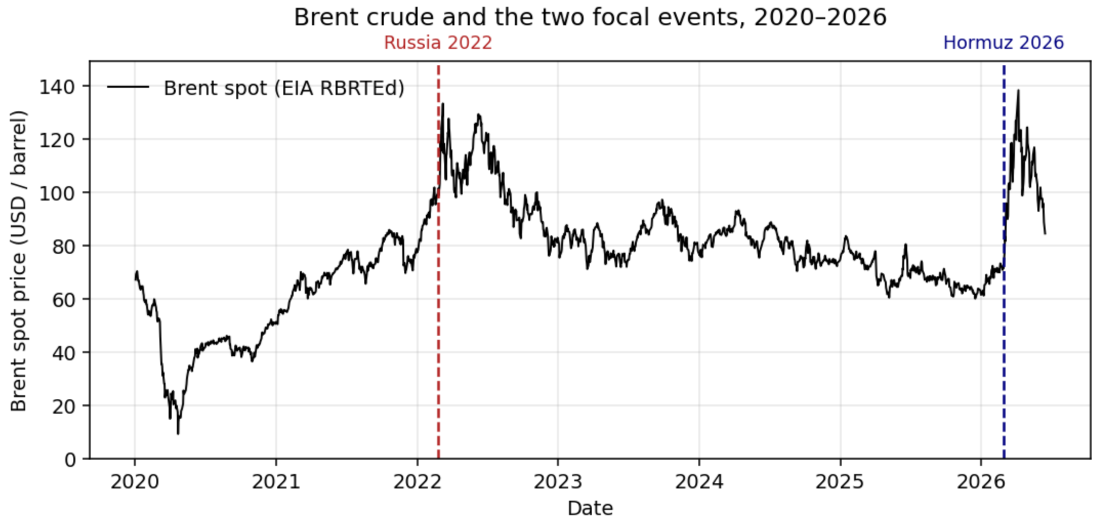

Figure 1: Brent crude spot price, 2020–2026, with the two focal event dates. Brent roughly doubled after both shocks from a similar ∼\$75 base, but from very different pre-event trajectories, a steep 2020–22 recovery rally before Russia and a mild net decline through 2024–25 before Hormuz. This trajectory asymmetry is what makes a naïve trend-extrapolation counterfactual unreliable (Section 6).

5 Empirical Strategy and Validation

5.1 Estimators

All five estimators (Table 1) take the same input (log donor levels) and target log Brent over the pre-window, then project the post-window counterfactual. The two events are estimated entirely separately: they share the same 19-donor pool and the same five-estimator pipeline, but each event trains its own models on its own pre-window, so no parameter learned on one event is carried to the other. *Convex SCM* solves the classic nested optimisation: non-negative weights summing to one, chosen to minimise pre-period RMSPE (Abadie et al., 2010). *Augmented SCM* adds a ridge penalty that corrects for imperfect pre-fit and allows limited extrapolation (Ben-Michael et al., 2021). The *elastic net* relaxes the simplex to signed, sparse coefficients (Zou and Hastie, 2005; Doudchenko and Imbens, 2016). *XGBoost* fits shallow gradient-boosted trees (Chen and Guestrin, 2016), deliberately constrained (depth-2 stumps, strong regularisation, row/column subsampling) for the small-sample regime. *Bayesian ridge* is a conjugate Bayesian linear regression on donor levels, the regression component of BSTS (Brodersen et al., 2015). Hyperparameters are tuned over small grids (3–9 configurations) to bound the selection bias of Cawley and Talbot (2010); the convex SCM has no tuned hyperparameter. Because the five estimators produce incommensurable weight vectors, a simplex for convex SCM, signed unconstrained coefficients for the regressions, and no linear weights at all for XGBoost, the ensemble operates on the *gap series* each model implies rather than on the weights. The headline at every t is the cross-model median gap, with the inter-quartile range as the model-uncertainty band.

5.2 Validation battery and its logic

A credible SCM estimate must clear three levels (Abadie, 2021): pre-event fit, donor identification, and inference. We organise the validation around these three levels and run every test per model and per event.

The first level concerns *fit*. Out-of-sample tracking is assessed by expanding-window walk-forward cross-validation (Hyndman and Athanasopoulos, 2018; Bergmeir and Benítez, 2012). Its role here is not model selection but evidence against over-fitting: a synthetic whose out-of-sample pre-period tracking matches its in-sample tracking is reproducing a stable donor–Brent relationship rather than memorising the pre-window, which is the empirical content of the common-factor spanning assumption.

The second level concerns *identification*, and is the donor-cleanliness battery that defends the no-interference (SUTVA) assumption. Before any model is fit, each donor's own return series is tested for a mean shift at T_0 using a permutation form of the known-break test of Chow (1960) (rather than the unknown-break test of Andrews (1993), since the date is known), together with a Wilcoxon event-window test (Brown and Warner, 1985); a donor is flagged if either rejects after Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995). Regime stability is checked with distance correlation across pre-period thirds (Székely et al., 2007) and with Bai–Perron break dating (Bai and Perron, 1998, 2003), and an I(0)/I(1)-robust trend-break test (Harvey et al., 2009; Kwiatkowski et al., 1992) confirms that no donor's own trend breaks at either T_0; the single VIX relationship-break noted in Section 4 falls ∼7 weeks *before* the Russia T_0, not at it, and does not recur on the analysis window. Empirically, for Hormuz 31 of 32 donors agree between audit and tests (only Cotton differs); for Russia the tests confirm the broad picture and add one flag (CNY) attributable to concurrent China dynamics rather than the war.

The third level concerns *inference*. We run the in-space placebo, refitting each donor as a placebo treated unit and ranking Brent's post/pre RMSPE ratio against the resulting distribution (Abadie et al., 2010) and reading the permutation p-value with the symmetry caveat of Hahn and Shi (2017). We add the in-time placebo at a fake T_0 six months early, formalised as the mixed placebo test of Chen and Yan (2023), and a leave-one-donor-out check. Separately, as an exploratory probe of regime stability rather than a validation test, we apply the weights fitted on one event to the other event's window and read how far the pre-period fit degrades. We report this transfer in Section 6 for what it reveals about how regime-bound the factor structure is, not as evidence for the headline magnitude, which rests entirely on the separately fitted within-event ensemble.

6 Results

6.1 Pre-event fit

Table 5 reports walk-forward cross-validation. The pattern matches the two events' pre-periods. For Hormuz, whose pre-period is calm, four of five models clear the heuristic overfit flag and the fifth sits only just past it. For Russia, whose pre-period spans the volatile 2020–22 recovery, more models trip the flag, because a noisy pre-period inflates validation error. The flags do not exclude any model. They mark which synthetics to read with more caution, and the most flagged model is also the one whose headline gap is the ensemble outlier, so the flag and the outlier tell a consistent story.

6.2 Headline estimates

Table 6 reports the per-model post-window gap and the drift-adjusted gap for both events, and Figure 2 plots the observed and synthetic paths. For **Russia 2022** the ensemble-median premium is 22.0 % over the seven-month window (IQR 10.9 % to 30.8 %), which implies a counterfactual mean of \$88.6/bbl against an actual \$108.5. This is consistent with the documented \$95→\$130 move and its partial reversion by September. For **Hormuz 2026** the ensemble-median premium is 55.9 % over the window to mid-June (IQR 51.2 % to 56.6 %), with an actual post-window

Table 5: Walk-forward cross-validation (5 expanding folds, 20-day horizons). A validation/training RMSE ratio above 2 is a heuristic overfit flag, not an exclusion rule.

Event	Model	val/train RMSE	Flag
Russia	Convex SCM	1.25	OK
Russia	Augmented SCM	2.06	overfit
Russia	Elastic net	1.99	OK
Russia	XGBoost	3.24	overfit
Russia	Bayesian ridge	3.09	overfit
Hormuz	Convex SCM	1.40	OK
Hormuz	Augmented SCM	1.54	OK
Hormuz	Elastic net	1.51	OK
Hormuz	XGBoost	1.84	OK
Hormuz	Bayesian ridge	2.28	overfit

mean of \$106.9 against an implied counterfactual near \$68.6. The Hormuz band is far tighter than Russia's, about 5 points against about 20, which reflects the calmer Hormuz pre-period.

Table 6: Headline post-event gaps (%). Drift-adjusted = raw − drift contribution. Ensemble = cross-model median, band = IQR (Q25–Q75).

	Russia 2022 (7 mo)		Hormuz 2026 ($\sim$3.5 mo)	
Model	Raw	Adj.	Raw	Adj.
Convex SCM	33.6	28.4	55.9	57.3
Augmented SCM	10.9	10.6	56.8	57.1
Elastic net	22.0	21.2	56.6	57.4
XGBoost	30.8	28.9	51.2	52.7
Bayesian ridge	−9.0	−9.1	43.0	43.0
Ensemble median	**22.0**	**21.2**	**55.9**	**57.1**
IQR	**[10.9, 30.8]**	n/a	**[51.2, 56.6]**	n/a

Economic reading of the model spread. The cross-model dispersion has an economic explanation rather than reflecting noise, and this is the one place where the individual models matter, so we name them here. Convex SCM, confined to the simplex, leans on the most stable donors, the FX block, as a trend baseline and adds soft-commodity deltas such as Coffee and Sugar for Russia. The Bayesian ridge, with unconstrained Gaussian coefficients, loads heavily and with large opposing signs on currencies. For Russia those currencies moved with the simultaneous and historically large Fed tightening cycle, a macro factor that has nothing to do with the invasion. That confound is why the Bayesian ridge becomes the Russia outlier, with a negative full-window gap, and why it carries the highest walk-forward flag. It is the expected behaviour of an unregularised linear model in a pre-period that contains a strong non-event driver, and it is the reason the ensemble uses the median rather than the mean. For Hormuz, whose calmer pre-period has no analogous confound, the five models agree to within about 5 points.

6.3 Inference: the in-space placebo

Table 7 and Figure 3 report the in-space placebo. For **Hormuz** the result is as strong as the test allows. Brent ranks first among all placebo units under every model and both donor pools, with a post/pre RMSPE ratio between 6.9 and 10.2 that exceeds every donor's, so the permutation

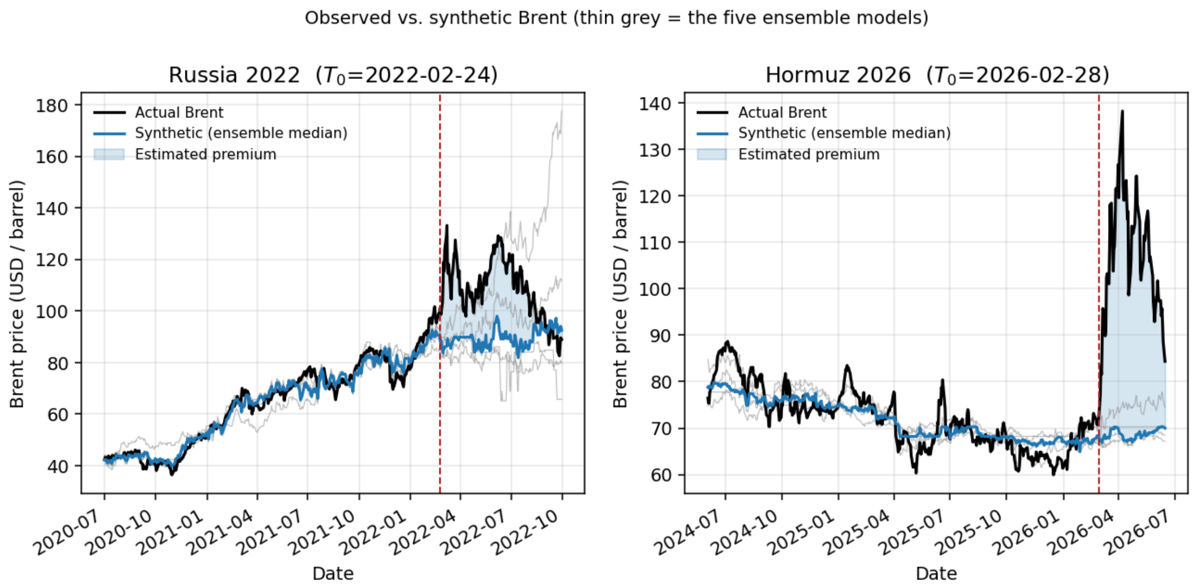

Figure 2: Observed Brent (black) vs. the synthetic counterfactual. The thick blue line is the ensemble-median synthetic, thin grey lines are the five individual models, and the shaded region after T_0 (dashed red) is the estimated premium. The synthetics track Brent closely before the event and decouple sharply afterward.

p-value sits at the attainable floor, 0.050 in the 19-donor pool and 0.030 in the 32-donor pool. For **Russia** no model reaches the floor, and XGBoost comes closest at $p = 0.20$. The reason is mechanical. The volatile 2020–22 pre-period gives the donors large pre-period RMSPE as well, which widens the placebo distribution and lowers the test's power. This is a known low-power property of the in-space placebo under a volatile pre-period (Abadie et al., 2010), not evidence of a small effect, and we validate Russia's magnitude independently below.

Table 7: In-space placebo permutation p-values and Brent's post/pre RMSPE ratio (19-donor pool). Floor = $1/(N+1)$: 0.050 (19-donor), 0.030 (32-donor).

Event	Model	p (19-donor)	p (32-donor)	Brent ratio
Russia	best of five (XGBoost)	0.20	0.15	7.75
Hormuz	Convex SCM	**0.050**	**0.029**	6.93
Hormuz	Augmented SCM	**0.050**	**0.029**	8.45
Hormuz	Elastic net	**0.050**	**0.029**	8.53
Hormuz	XGBoost	**0.050**	**0.029**	10.19
Hormuz	Bayesian ridge	**0.050**	**0.029**	10.24

6.4 Robustness

Russia magnitude and external validation. Because Russia is a known event, we validate its magnitude against the EIA's last two pre-invasion Short-Term Energy Outlook forecasts, issued before 24 February 2022. These forecasts are an independent structural counterfactual that shares no information path with the SCM. Over the identical 151-day window the ensemble-median synthetic mean of \$88.6 sits within \$3.6 of the STEO February 2022 forecast of \$84.9, and three of five models bracket the STEO anchor to within \$4. Two independent methods landing on nearly the same counterfactual is strong evidence that the SCM does not produce an implausible path.

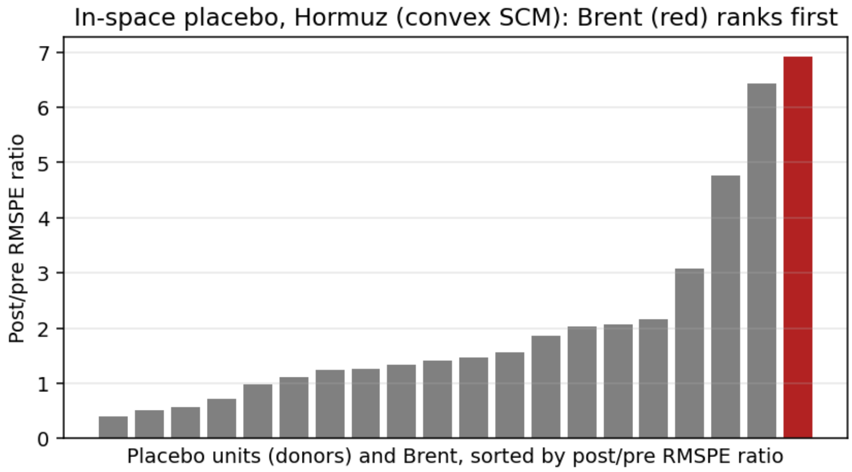

Figure 3: In-space placebo for Hormuz (convex SCM). Each bar is one unit's post/pre RMSPE ratio. Brent (red) is the largest and ranks first among all 19 placebo units, so the permutation test rejects at the attainable floor.

Post-window horizon sensitivity. Figure 4 reports the gap at 1, 2, 3, and 6-month and full horizons. The two events show opposite and diagnostic profiles. Russia declines with horizon, from about 31% to 22%, which is the signature of delayed contamination as second-round war channels switch on through 2022 and pull the synthetic up. The full-window figure is therefore conservative, and the least-contaminated one-month estimate is about 31%. Hormuz rises and then eases, from about 46% to 59% and back to 56%, which is the opposite of attenuation. The chokepoint premium builds in and holds, and the slight easing at the full horizon reflects Brent's partial mid-June pullback that appears once we extend the window to the latest data. The same diagnostic declining for Russia and rising for Hormuz shows that it is not a mechanical artefact.

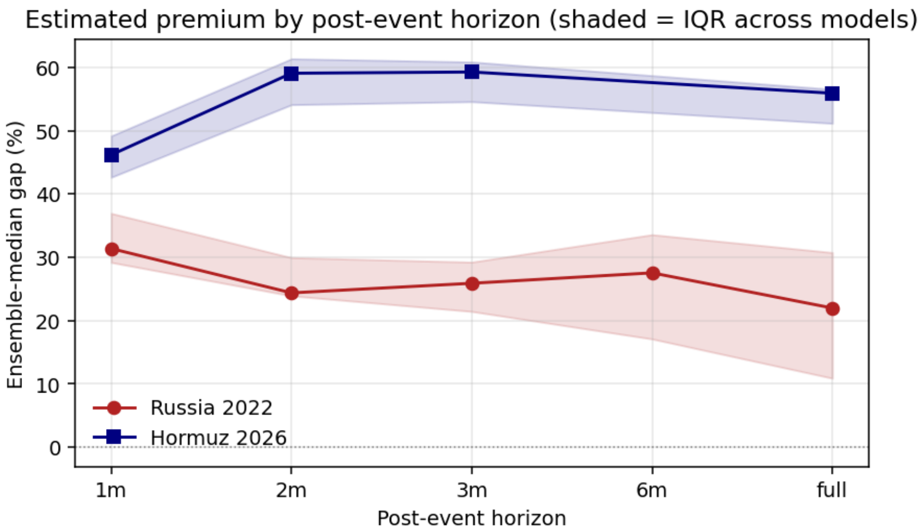

Figure 4: Ensemble-median gap by post-event horizon (shaded = IQR across models). Russia attenuates with horizon, the delayed-contamination signature, while Hormuz rises and then plateaus as the premium builds in and holds.

Leave-one-donor-out. Dropping each weighted donor in turn leaves the focal Hormuz estimate tight around its baseline across all models (convex 51.2 to 59.3, ASCM 54.6 to 60.5, elastic 55.2

to 62.2, XGBoost 51.6 to 57.7%), so no single donor drives the result. For Russia we report the ensemble-median range rather than every model, because Russia's wider spread comes from the simplex models' reliance on a few soft-commodity donors, the same concentration that drives the contamination finding below. Read that way, the wider Russia range (for example convex 32.8 to 59.6%) is itself a symptom of the donor-quality issue rather than a separate concern.

Cross-event weight stability. As an exploratory probe rather than a validation test, we apply the weights fitted on one event to the other event's window and read how far the pre-period fit degrades. The transfer is weak, and that weakness is informative. Only convex SCM transfers to a clearly positive Hormuz premium, at 25.3% with a degraded but non-pathological pre-fit, while the other models move toward or below zero with pre-fit errors many times their independent level (Table 8). The reason is that the two pre-periods are different factor regimes. Russia's 2020–22 pre-period is shaped by the COVID recovery and the onset of Fed tightening, whereas Hormuz's 2024–26 pre-period is a calmer disinflation regime, so the donor-to-Brent loadings that the weights encode do not carry across. The models that fail hardest are the ones that load most on currencies, and the dollar and emerging-market FX relationships shifted most between the two regimes. Extending the Hormuz window through the June pullback widens the regime gap further. We therefore read the transfer as weak and convex-only corroboration that some positive premium survives a change of regime, and as direct evidence that fitting each event separately is the correct choice, not as confirmation of the level.

Table 8: Cross-event weight stability probe: Russia-fitted weights applied to Hormuz. Independent and transferred pre-period RMSPE in log units.

Model	Indep. gap (%)	Transf. gap (%)	Indep. RMSPE	Transf. RMSPE
Convex SCM	55.9	25.3	0.066	0.336
Augmented SCM	56.8	−0.8	0.066	0.467
Elastic net	56.6	−68.6	0.054	1.513
XGBoost	51.2	−44.1	0.053	0.928
Bayesian ridge	43.0	−100.0	0.037	18.567

Naïve counterfactual floor. We compare the SCM against donor-free univariate counterfactuals built only from Brent's own past (Table 9), and the comparison yields three findings. First, the SCM adds value. It sits about 13 points above the trend-free random-walk null for Russia and about 7 points above it for Hormuz, so the donor machinery extracts something that extrapolating Brent's own series cannot. Second, and most telling, the naive bias flips sign across the two events purely because the pre-window trend sign differs. The linear trend extrapolates Russia's recovery rally upward and returns an implausible negative full-window gap of about −4%, yet the same method extrapolates Hormuz's mild downtrend and inflates the gap to about 71%. A counterfactual built from one series alone is therefore regime-fragile in a way the donor-based SCM is not. Third, for Russia the SCM and the trend extrapolation sit on opposite sides of the truth. The SCM leans high, as the contamination analysis below confirms, and the trend extrapolation leans low, so the two bracket the real value, with the random-walk null at +8.7% the most defensible naive point. The naive floor and the contamination analysis thus corroborate each other, one from a donor-free direction and one from a donor-based one.

Sign of the contamination bias. This last check is a deliberately naive and directional diagnostic, not a structural correction. It does not prove that the donors are clean. It asks a narrower question. If some contamination remains despite the audit and the break tests, which way would it bias the estimate? We answer it with equation (3) and a Brent-free reference, and

Table 9: SCM vs. naïve univariate counterfactuals, full-window gap (%). RW flat and RW drift are the random-walk carry and the drifted random walk, Lin. trend is the linear pre-trend extrapolation, and ARIMA is the AIC-selected model, which reduces to a driftless random walk for both events.

Event	SCM median	RW flat	RW drift	Lin. trend
Russia	22.0	8.7	−6.9	−4.2
Hormuz	55.9	48.9	49.8	70.9

the two events give opposite and informative answers. For **Hormuz**, where every other check already supports a stable estimate, any residual contamination has a small negative sign, between about −1 and −8%. The reading is reassuring in one direction. If the estimate is biased at all, it is biased downward, so the true premium is at least as large as we report and the headline is conservative. For **Russia** the sign reverses and the bias is large and positive, on the order of +13 to +21%. The reason is economic. The high-price and risk-off environment of 2022 depressed the demand-cyclical donors, the equities, metals, and soft commodities, relative to the safe-haven and currency donors, so the synthetic built from them sits too low and the gap reads too high. The bias concentrates in the low-reliability soft-commodity donors, and splitting donors by pre-window factor R^2 confirms that most of it sits in the low-reliability group rather than in the donors that track the common factor well. The conclusion is asymmetric. Contamination is conservative for the focal Hormuz estimate and inflates the Russia full-window number, so Russia's magnitude validation rests on the early and peak window, where the \$95→\$130 move is recovered, and not on the attenuated full-window average.

6.5 Interpretation

The 2026 Hormuz chokepoint premium is roughly 56 % of Brent's no-event level over the first three and a half months, and the estimate is stable. It holds across all five estimators and clears the validation battery of Section 5, with the in-space placebo reaching its attainable floor as the strongest single result. The contamination analysis adds a useful direction to this stability. If anything, the true Hormuz premium is larger than our estimate, not smaller. The cross-event probe is the weakest of the checks, and we lean on it only for the sign of the effect under a change of regime, never for its level. We read the estimate as a complement to the structural scenario estimates from energy institutions rather than a competitor to them. Those answer the question of what price an assumed barrel shortfall implies, while the SCM answers what price gap the observed cross-asset co-movement implies, and the two questions inform different parts of a policy response. For the levers set out in Section 1, reserve releases, monetary pass-through, and fiscal buffers, the relevant input is the integrated path-level premium, and the estimate here supplies it together with an auditable identification argument.

6.6 Limitations

Four limitations bound what we can claim. First, the design has a single treated unit and a short post-period, which caps the power of permutation inference. The Hormuz placebo reaches 0.050, which is the best value attainable at this sample size rather than a conventional rejection, so we read it as the strongest available signal and not as a small p-value in the usual sense. Second, the estimand is a reduced-form total effect. It captures the full price gap but does not separate the premium into a physical-supply component, a risk-premium component, and an expectations component, which some policy uses would want to distinguish. Third, the Hormuz event is recent and has no documented post-event counterfactual of its own, so its validation is internal and rests on the Russia case as a calibration by analogy rather than on an external ground truth. Fourth, the factor structure is regime-bound, as the weak cross-event transfer shows directly. The estimate is valid for the regime of this event and should not be read as a general chokepoint premium that would recur unchanged in a different macro environment. Underlying all four is that donor cleanliness is only partly testable. The no-interference assumption concerns an unobserved counterfactual donor path, and the contamination-sign diagnostic is naive and directional, so it bounds the likely direction of any residual bias without proving that none exists.

7 Conclusion

We estimated the Brent-price effect of the 2026 Strait of Hormuz disruption by building a counterfactual with an ensemble synthetic control and benchmarking the design against the documented Russia 2022 shock. Over the window to mid-June 2026 the ensemble-median premium is roughly 56 % of Brent's no-event level (IQR 51 to 57%), large and positive under every estimator. The estimate clears the validation battery. Brent ranks first among all units in the in-space placebo at the attainable floor, the leave-one-out check stays tight, the contamination-bias sign points conservative, and the synthetic beats the naive univariate baselines. On the benchmark event the same machinery recovers a magnitude consistent with the documented \$95 to \$130 move and within \$3.6 of the EIA's pre-invasion structural forecast, which gives independent reassurance that the method tracks a real path.

The contribution is twofold. The substantive result is a data-driven Brent premium for the Hormuz disruption that complements the structural scenario estimates produced by energy institutions. The methodological result is a transparent template for validating cross-asset synthetic control under a single-unit geopolitical treatment, with an identification argument a reader can audit donor by donor. Our approach and the structural one answer different questions. The structural models ask what price an assumed barrel shortfall implies, while the synthetic control asks what price gap the observed co-movement of global markets implies. Because the two read the event from different directions, they work best together, and our estimate is a complement to scenario analysis rather than a substitute for it.

For policy the relevant object is the integrated, path-level premium rather than a single peak-day figure, and that is what the estimate supplies. Reserve sizing, the monetary decision to look through or tighten, fiscal buffering in oil importers, and sovereign hedging all act on the magnitude and persistence of the price move, and a counterfactual path lets a decision maker read both at once and update as new data arrive.

Several limitations bound the estimate. The design has a single treated unit and a short post-window, so the inference tests are under-powered and the placebo can only reach its floor. The estimand is a reduced-form total effect and does not separate supply, risk, and expectations. The Hormuz event is recent and has no external counterfactual of its own, so its validation is internal and leans on the Russia benchmark by analogy. The factor structure is regime-bound, as the weak cross-event transfer shows. None of these overturns the central finding. The observed co-movement of global markets with Brent implies that the 2026 Hormuz disruption added a large and robust premium to crude, on the order of half its no-event price, and the estimate is meaningful and usable for the decisions that depend on it.

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A Robustness grid

The headline uses the preferred specification (shared 19-donor pool, preferred pre-window, ensemble median). The full grid varies (i) the pre-window (preferred, extended, and narrow specifications); (ii) the donor pool (shared 19, permissive 26 for Russia, full 32); and (iii) the model (each of the five plus the median). For Russia the donor-pool sweep is itself a diagnostic: the gap is expected to *shrink* monotonically from the clean 19-donor pool to the contaminated 32-donor pool, because Russia-treated donors get inflated by the same shock and pull the synthetic up. If this monotonic pattern holds, the SUTVA-driven exclusion logic is empirically supported.

B Bayesian ridge and full BSTS

The fifth ensemble member implements only the regression-on-donors term $\beta' x_t$ of the Bayesian structural time-series model of Brodersen et al. (2015), with a conjugate Gaussian–Gamma prior. It omits the local-linear *trend*, the *seasonal* component, the *spike-and-slab* donor selection, and the autocorrelation-aware posterior over the full state trajectory. At our sample size, with no strong weekly seasonality in daily oil prices, the regression term carries most of the signal, so the proxy is an adequate Bayesian-flavoured diversifier; but its credible intervals are Gaussian and ignore autocorrelation, and its donor "importances" are standardised-coefficient magnitudes, not true posterior inclusion probabilities. We therefore call it a Bayesian ridge regression rather than a BSTS, and read its placebo p-values as rank statistics rather than formal probabilities (Table 7). A full BSTS implementation is the natural route to autocorrelation-aware credible intervals in a future iteration.

C Structural-break tests

Two break tests support donor screening. *Bai–Perron* (Bai and Perron, 1998, 2003) dates breaks in the donor–Brent weekly-return relationship $r_t^{\text{donor}} = \alpha_s + \beta_s r_t^{\text{Brent}} + \varepsilon_t$ by globally minimising segment SSR via dynamic programming, with the break count chosen by BIC. Only VIX shows a break strictly inside a pre-window (2022-01-07). We treat the break tests as informative rather than automatically disqualifying: the VIX break is a loading-stability signal, it does not reproduce on the analysis window alone, and the in/out sensitivity is below 1 pp, so VIX is retained (Section 4). The relationship specification is Brent-based, so a break *at* T_0 is ambiguous; the complementary Harvey et al. (2009) trend-break test resolves this by testing the donor's *own* log-level trend with a statistic whose asymptotic size is invariant to whether the series is I(0) or I(1): it weights an I(0)-optimal and an I(1)-optimal sup-t statistic by KPSS stationarity weights (Kwiatkowski et al., 1992). Empirically all 32 donors are classified I(1)-like and none rejects a trend break within ± 8 weeks of either T_0, so no retained donor was itself trend-treated by either event.